SIMATS ENGINEERING

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA12	**Course Title:**	Computer Architecture
CO Assessed:	CO2 – Precision (BL3)	**Assessment:**	Analytical Problem Solving
Weightage:	45%	**Total Marks:**	100
CO2 Statement:	Apply integer and floating-point arithmetic algorithms including Booth's multiplication, restoring/non-restoring division, and IEEE 754 standard operations to solve fixed-point and floating-point computational problems.		
Student Name:		**Reg. No.:**	
Section / Batch:		**Date:**	

Instructions to Students

- Identify the appropriate arithmetic algorithm or number representation required for the given computational problem.
- Analyse the step-by-step execution of integer and floating-point arithmetic operations using the selected algorithm.
- Apply Booth's multiplication, restoring/non-restoring division, and IEEE 754 standards to obtain accurate results.
- Compute binary multiplication, division, normalization, and floating-point arithmetic using appropriate algorithms.
- Interpret the computed results by evaluating their accuracy, precision, and suitability for the given application.

QUESTIONS

1. A processor performs arithmetic operations on signed 8-bit integers using 2's complement representation. Consider two numbers: +45 and −23. Analyse in detail how the system performs both addition and subtraction, including binary conversion, alignment of sign bits, and intermediate steps. Determine whether overflow occurs, and explain how the processor detects overflow and interprets the final result.
2. A high-performance computing system replaces a ripple carry adder with a 4-bit carry look-ahead adder (CLA) to reduce computation delay. Given two binary inputs, analyze how generate and propagate signals are formed and how carries are computed in advance. Compare the time delay of CLA with ripple carry addition and justify, with reasoning, why CLA improves speed in modern processors.
3. In a signed multiplication operation, a processor uses Booth's algorithm to multiply two numbers, −6 and +5. Analyze the algorithm step-by-step, including the role of the accumulator, multiplier, and Booth bit, along with arithmetic shifts. Explain how the algorithm efficiently handles signed numbers and reduces the number of addition/subtraction operations compared to traditional multiplication.
4. A processor is required to perform division of two binary numbers (e.g., 13 ÷ 3) using both restoring and non-restoring division algorithms. Analyze both methods step-by-step, showing intermediate remainders and quotient formation. Compare the two approaches in terms of number of steps, complexity, and efficiency, and explain why non-restoring division is often preferred in modern systems.
5. A scientific application requires precise handling of real numbers using the IEEE 754 standard for floating-point representation. Given a decimal number (e.g., −13.25), analyse how it is represented in both single precision and double precision formats, including sign, exponent, and mantissa fields. Further, explain how floating-point addition is performed between two such numbers, highlighting issues like normalization, rounding, and precision loss.

Code of Conduct

I __________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _________________________

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Marks Evaluation:

Q. No	Problem Understanding (20)	Method Selection (20)	Solution Process (25)	Accuracy of Calculation (20)	Interpretation of Results (15)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 20	/ 25	/ 20	/ 15	/ 100

Course Coordinator **Faculty Incharge**

13/8/26

Assessment Tool - 1
Analytical Problem Solving

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1)

2's complement :

(i) Addition :

(+45) + (-23)

+45 = 00101101

+23 = 00010111

1's = 11101000

+1

2's = 11101001

∴ -23 = 11101001

00101101
+ 11101001

MSB $\boxed{1}$ 00010110

∴ 00010110

45 + (-23) = 22

(ii) Subtraction ; (+45) - (-23)

+45 = 00101101

-23 = 11101001

-23 = 1's = 00010110

+1

2's = 00010111

+45 = 00101101

-23 = 00010111

01000100

$+45-(-23) = 01000100$

$\therefore +45-(-23) = 68$ //.

2) 4 - bit Carry Look - Ahead Adder (CLA)

$A = 1011$, $B = 0110$, $C_{in} = 0$

Arranging bits:

		A	B
A3	B3	1	0
A2	B2	0	1
A1	B1	1	1
A0	B0	1	0

$G_i = A_i \times B_i$

$G0 = A0 \times B0 = 1 \times 0$, $G0 = 0$

$G1 = A1 \times B1 = 1 \times 1$, $G1 = 1$

$G2 = A2 \times B2 = 0 \times 1$, $G2 = 0$

$G3 = A3 \times B3 = 1 \times 0$, $G3 = 0$

$\therefore$ $P_i = A_i \oplus B_i$

$P0 = 1 \oplus 0 = 1$

$P_1 = 1 \oplus 1 = 0$; $P2 = 0 \oplus 1 = 1$

$P3 = 1 \oplus 0 = 1$

$\therefore$ To find carry $C0 = 0$, $C1 = G0 + P0C0$

$C1 = 0 + (1 \times 0) = 0$

$C2 = = 1 + 0 \times 0 = 1$

$C3 = 0 + 1 \times 1 = 1$

$C4 = 0 + 1 \times 1 = 1$

To find sum:

$S_i = P_i \oplus C_i$

$S_0 = P_0 \oplus C_0$

$= 1 \oplus 0 = 1$

$S_1 = 0 \oplus 0 = 0$

$S_2 = 1 \oplus 1 = 0$ $S_3 = 1 \oplus 1 = 0$

Carry $= C_4 = 1$ $\therefore$ Sum $= S_3S_2S_1S_0$

$= 0001$

Final Answer: $= 10001_2$

3) Booth's Algorithm -6 and $+5$

$\underset{(M)}{-6} \times +5 \rightarrow$ Multiplier (Q)

n	A	Q	q_0	Action
4	0	0101	0	Initialization
	1010	0101	0	A = A − M
	0011	0010	1	ASR
3	0011	0010	1	A = A + M
	1110	1001	0	ASR
2	1110	1001	0	A = A − M
	0010	0100	1	ASR
1	0010	0100	1	A = A + M
	1110	0010	0	

$AQ = 11100010_2$

$-6 \times (+5) = -30$

4) Non-Restoring Division Algorithm 13 ÷ 3

n	A	Q	Action
4	0000	1101	Initialization
	0001	1010	LS(A, Q)
	1110	1010	A = A - M
	1110	1010	set Quotient bit = 0
3	1101	0100	LS(A, Q)
	0000	0100	A = A+M
	0000	0101	set Quotient bit = 1
2	0000	1010	LS(A, Q)
	1101	1010	A = A - M
	1101	1010	set quotient bit = 0
1	1011	0100	LS(A, Q)
	1110	0100	A = A+M
	0001	0100	set Quotient bit = 1

(Q) Quotient = 4 ; Remainder (A) = 1

5) 13.25 to binary :

i) 13 = 1101

0.25 = 0.01

13.25 = 1101.01_2 (-ve)

ii) $1101.01 = 1.10101 \times 2^3$

iii) sign = 1(-ve)
(1 bit)

Exponent (8 bits) = actual bit + bias
= (127)

= 3 + 127

mantissa (23 bit) = 130 = 1000010

10101000000000000000000

iv) (64-bit) Sign = 1

Exponent (11 bits) = 3 + bias (1023)
= 1026 = 1000000010

mantissa = 10101 followed by zero to 52 bits double precision to 52 bit

Double precision bit pattern :

1 1000000010

v) (-13.25 + 5.75) float pt addition

i) $1.0111 \times 2^2 = 0.10111 \times 2^3$

ii) $1.10101 - 0.10111 = 0.11110$

iii) $= -0.11110 \times 2^3$ (iv) -1.1110×2^2

v) $-13.25 + 5.75 = -7.5$